Multi-Agent System Designer

Design Multi-Agent Systems, Intelligently

APEX guides you through a structured interview to architect production-ready multi-agent systems. Seven proven SMA patterns. Self-improving RAG. Dual output.

Start Designing → View Patterns

CAPABILITIES

Everything you need to architect multi-agent systems

7 SMA Patterns

Proven multi-agent architectures from academic research. Sequential, parallel, hierarchical, debate, reflection, router, and custom hybrid patterns.

Self-Improving RAG

Retrieval-Augmented Generation that learns from every design session. Pattern knowledge evolves with your usage, getting smarter over time.

Dual Output

Generate both complete SMA architectures and reusable Claude Code skills. Export production-ready agent configurations and orchestration code.

APEX

- Projects
- New Session
- Patterns
- Skills
- Monitor

Projects

Search projects...

Total Projects	Patterns Used	Skills Generated	Active Agents
12	5	47	3

Recent Projects

View all

NutriCoach AI

Hierarchical

Multi-agent nutrition coaching system with 4 specialized agents

4 agents 2 days ago

FlowLearn Platform

Router

Adaptive learning platform with content routing and assessment agents

6 agents 5 days ago

CodeReview Bot

Sequential

Automated code review pipeline with linting, security, and style agents

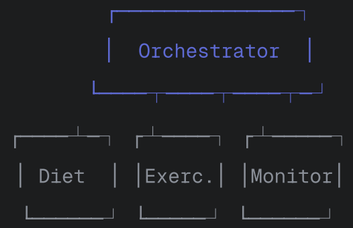
3 agents 1 week ago

A

APEX Architect

I've analyzed your project description. NutriCoach AI is a nutrition coaching platform that needs multiple specialized agents working together. Let me walk you through the architecture design.

Recommended Pattern: Hierarchical



A supervisor agent orchestrates 3 specialized sub-agents, each handling a distinct nutrition domain.

A

APEX Architect

How should the agents communicate with each other? This affects latency, reliability, and debugging complexity.

Choose a communication pattern:

- **Direct Message Passing** — Agents send structured messages directly. Low latency, simple to debug.
- **Shared State (Blackboard)** — Agents read/write to shared memory. Good for collaborative tasks.
- **Event-Driven (Pub/Sub)** — Async events via message broker. Best for decoupled, scalable systems.
- **Hybrid (Mixed)** — Combine patterns per agent pair. Maximum flexibility, higher complexity.

Type a message or provide additional context...

Agent Specification

Orchestrator Agent

ROLE

Coordinates all sub-agents, manages task delegation and result aggregation

TOOLS

`task_router` `state_manager`

Diet Specialist Agent

ROLE

Analyzes nutritional needs, creates meal plans, tracks dietary goals

TOOLS

`nutrition_db` `meal_planner`

DESIGN PROGRESS

2 of 7 phases complete

APEX

Search patterns...

Pattern Library

7 proven multi-agent system architectures backed by academic research

All Patterns | Sequential | Parallel | Hierarchical | Debate | Reflection

Sequential Chain

Pipeline

Agents execute in a fixed order, each passing output to the next. Ideal for deterministic workflows.

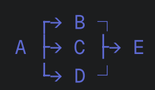


Wu et al. 2023 · AutoGen

Parallel Fan-Out

Concurrent

Multiple agents work simultaneously on sub-tasks. A reducer aggregates results.



Li et al. 2024 · LangGraph

Hierarchical

Supervisor

A supervisor orchestrates specialist sub-agents. Best for complex, multi-domain systems.

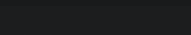


Hong et al. 2023 · MetaGPT

Multi-Agent Debate

Adversarial

Agents argue opposing positions to find better solutions through dialectical reasoning.

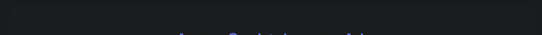


Du et al. 2023 · MAD

Self-Reflection

Iterative

An agent reviews and improves its own output through multiple self-critique iterations.

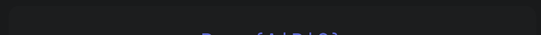


Shinn et al. 2023 · Reflexion

Router / Dispatcher

Dynamic

A router classifies input and delegates to the best-fit specialist agent dynamically.

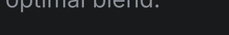


Wang et al. 2024 · AgentScope

Custom Hybrid

Composite

Combine multiple patterns to match complex real-world requirements. APEX generates the optimal blend.



APEX Custom · Adaptive

Project History

12 projects

Name	Pattern	Agents	Status	Date
NutriCoach AI	Hierarchical	4	Completed	Apr 3, 2026
FlowLearn Platform	Router	6	Completed	Mar 29, 2026
CodeReview Bot	Sequential	3	Completed	Mar 21, 2026
Customer Support SMA	Debate	5	In Progress	Apr 5, 2026
Research Assistant	Reflection	2	Completed	Mar 15, 2026
IoT Data Pipeline	Parallel	8	Draft	Apr 5, 2026